DIGITAL LIFESTYLE AND PRODUCTIVITY

A Data Visualization Study on Social Media Usage, Screen Time and Productivity Patterns

Course: Introduction to Data Analysis
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Submitted by:

1. Yuktha H	1RUA25SCS0120
2. Kanishka Gowda N	1RUA25SCS0037
3. Niharikaa Dhanya Kumar	1RUA25SCS0064
4. Navya TS	1RUA25SCS0062
5. Afreen Unnisa	1RUA25SCS0004

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ABSTRACT

This project explores the relationship between digital lifestyle indicators and productivity-related measures using a public dataset titled “Social Media vs Productivity.” The dataset contains 30,000 records and 19 attributes, including daily social media time, work hours per day, perceived productivity score, dataset-labelled actual productivity score, stress level, sleep hours and screen time before sleep.

The data were examined using descriptive statistics and four visualizations: a histogram, a bar chart, a scatter plot and a box plot. The average daily social media time in the median-imputed dataset was approximately 3.11 hours. The correlation between daily social media time and the dataset-labelled actual productivity score was approximately −0.010, indicating an almost negligible linear association in this dataset. Average productivity scores across job categories were also very close, while stress-level quartiles were identical across the job categories.

An important limitation is that the dataset is described in an external research source as synthetic or simulated rather than observations collected from actual participants. In addition, it contains work hours rather than study hours, and screen time before sleep rather than total daily screen time. Therefore, the results are treated as an exploratory analysis of the supplied dataset and are not presented as evidence about all students or the general population.

Keywords: Digital Lifestyle, Social Media, Productivity, Data Visualization, Exploratory Data Analysis, Synthetic Dataset.

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1. INTRODUCTION

Digital platforms have become an important part of everyday life and are widely used for communication, entertainment, learning and work. Social media provides opportunities for information sharing and interaction, but excessive or unplanned usage may also interrupt concentration and reduce the time available for important tasks.

The relationship between digital habits and productivity is not necessarily the same for everyone. It may depend on the purpose of usage, daily routines, workload, sleep and individual behaviour.

Data visualization helps identify patterns, compare groups and examine relationships between variables. This project uses a dataset containing digital lifestyle and productivity-related attributes to explore selected patterns. The analysis focuses on daily social media time, productivity scores and stress levels across different job categories.

2. PROBLEM STATEMENT

The project investigates whether the supplied dataset shows observable patterns between daily social media usage and productivity-related measures.

It also examines whether productivity scores and stress levels differ across job categories. The project considers the limitations of the available variables in relation to the original topic, which refers to study hours, total screen time and self-reported productivity.

3. AIM AND OBJECTIVES

Aim:

To explore and visualize relationships between digital lifestyle indicators and productivity-related measures using the supplied dataset.

Objectives:

- To inspect the structure, attributes and completeness of the dataset.
- To summarize selected numerical and categorical variables using descriptive statistics.
- To visualize the distribution of daily social media usage.
- To compare average productivity scores across different job categories.
- To examine the relationship between daily social media time and productivity scores.
- To compare stress-level distributions across job categories.
- To derive data-supported findings and practical recommendations.

4. SCOPE OF THE STUDY

The analysis is limited to the 30,000 records and 19 attributes in the supplied CSV dataset.

It covers daily social media time, work hours per day, perceived productivity, dataset-labelled actual productivity, stress level, sleep hours, screen time before sleep, job type and preferred social media platform.

The project does not directly measure academic grades, actual study hours, total daily screen time or independently verified productivity. It also does not establish cause-and-effect relationships.

5. DATASET DESCRIPTION

The dataset contains 30,000 rows and 19 columns. It includes numerical, categorical and Boolean attributes.

The original file contains missing values in seven numerical columns and no duplicate rows. The dataset was obtained from the Kaggle collection titled “Social Media vs Productivity.”

An external 2026 research preprint describes this dataset as synthetic or simulated. Therefore, the records should not be described as 30,000 real survey respondents.

Table 1: Overview of the Dataset

Dataset Property	Description
Dataset Name	Social Media vs Productivity
Source	Kaggle
Original Dimensions	30,000 rows × 19 columns
Numerical Attributes	14
Categorical/Object Attributes	3
Boolean Attributes	2
Duplicate Rows	0
Nature of Data	Synthetic/Simulated
Collection Period	Not documented
Sampling Method	Not applicable as a real-participant survey

5.1 SELECTED ATTRIBUTES

The following attributes were selected for the analysis based on their relevance to digital lifestyle and productivity.

Table 2: Selected Dataset Attributes

Attribute	Description	Data Type
daily_social_media_time	Daily time spent on social media, in hours	Numerical
work_hours_per_day	Number of hours worked per day	Numerical
perceived_productivity_score	Perceived productivity score	Numerical
actual_productivity_score	Dataset-labelled actual productivity score	Numerical
stress_level	Stress-level score	Numerical
sleep_hours	Number of hours of sleep	Numerical
screen_time_before_sleep	Screen time before sleeping, in hours	Numerical
job_type	Category of occupation	Categorical
social_platform_preference	Preferred social media platform	Categorical

Note: The dataset contains work hours rather than study hours. Similarly, screen time before sleep is not equivalent to total daily screen time.

6. DATA SOURCE AND METHODOLOGY

The project uses secondary data obtained from the Kaggle dataset "Social Media vs Productivity."

The CSV file was loaded into Python using the pandas library. The Jupyter Notebook was used to inspect the dataset, identify missing values, remove duplicates, perform data preprocessing, detect outliers, scale numerical attributes and generate visualizations.

The analysis involved the following stages:

1. **Data Collection:** Downloading the dataset from Kaggle.
2. **Data Loading:** Importing the CSV file into Python using pandas.
3. **Data Inspection:** Examining the dimensions, data types and missing values.
4. **Data Cleaning:** Removing duplicate records and handling missing values.
5. **Outlier Detection:** Applying the Interquartile Range method to selected numerical columns.
6. **Feature Scaling:** Applying MinMaxScaler to numerical attributes.
7. **Data Visualization:** Creating graphs to identify distributions, comparisons and relationships.
8. **Interpretation:** Analysing the visualizations and identifying key findings.

Table 3: Methodology and Tools

Stage	Method / Tool	Purpose
Data Loading	pandas read_csv()	Load the CSV dataset
Data Inspection	shape, info(), isnull().sum()	Understand dataset structure
Missing Values	Median and mode imputation	Handle incomplete records
Duplicate Removal	drop_duplicates()	Remove repeated records
Outlier Detection	IQR Method	Identify extreme values
Feature Scaling	MinMaxScaler	Rescale numerical attributes
Visualization	Matplotlib and Seaborn	Generate graphical representations

7. DATA PREPROCESSING

Data preprocessing was performed to improve the quality and consistency of the dataset before analysis.

The original dataset contained 30,000 records and 19 attributes. No duplicate rows were identified. However, missing values were present in seven numerical columns.

The notebook used median imputation for numerical attributes. Median imputation was selected because it is less affected by extreme values compared to mean imputation.

Categorical columns were set up for mode imputation if missing values were present. However, the original dataset did not contain missing values in its categorical columns.

Table 4: Missing Values in the Original Dataset

Column	Missing Values	Percentage
daily_social_media_time	2,765	9.22%
perceived_productivity_score	1,614	5.38%
actual_productivity_score	2,365	7.88%
stress_level	1,904	6.35%
sleep_hours	2,598	8.66%
screen_time_before_sleep	2,211	7.37%
job_satisfaction_score	2,730	9.10%
Total	**16,187**	**—**

After handling missing values, the dataset retained 30,000 records and contained no missing values.

The Interquartile Range (IQR) method was subsequently applied to the following columns:

- daily_social_media_time
- sleep_hours
- screen_time_before_sleep

The notebook's sequential IQR filtering reduced the dataset from 30,000 to 29,454 records, removing 546 records.

MinMaxScaler was then applied to numerical columns in a separate scaled dataset to bring numerical values to a common range.

Important: The four figures in this report reproduce the supplied notebook's plotting code and use the median-imputed dataset before outlier removal and scaling. They should not be interpreted as visualizations of the later filtered or scaled dataset.

8. EXPLORATORY DATA ANALYSIS

Exploratory Data Analysis (EDA) was performed to understand the distribution and characteristics of the numerical and categorical attributes.

Descriptive statistics such as mean, median, minimum and maximum values were calculated to summarize the selected numerical variables.

Table 5: Descriptive Statistics of Selected Numerical Variables

Variable	Mean	Median	Minimum	Maximum
Daily Social Media Time	3.105	3.026	0.000	17.973
Work Hours Per Day	6.991	—	—	—
Perceived Productivity Score	5.511	—	—	—
Actual Productivity Score	4.952	—	—	—
Stress Level	5.545	—	—	—
Sleep Hours	6.500	—	—	—
Screen Time Before Sleep	1.024	—	—	—

Note: The complete mean, median, minimum and maximum values should be taken from the final notebook output. Only values verified from the supplied analysis have been populated here.

The mean daily social media time was approximately 3.11 hours, while the median was approximately 3.03 hours.

The average work hours per day was approximately 6.99 hours. This variable represents working hours and must not be interpreted as study hours.

The average screen time before sleep was approximately 1.02 hours. This does not represent total daily screen time.

8.1 DISTRIBUTION OF RECORDS BY JOB TYPE

Table 6: Number of Records by Job Type

Job Type	Number of Records
Education	5,055
IT	5,026
Finance	5,017
Student	5,012
Unemployed	4,958
Health	4,932
Total	**30,000**

The dataset contains records across six job categories, with relatively similar numbers of observations in each category.

8.2 DISTRIBUTION BY PREFERRED SOCIAL MEDIA PLATFORM

Table 7: Preferred Social Media Platforms

Platform	Number of Records
TikTok	6,096
Telegram	6,013
Instagram	6,006
Twitter	5,964
Facebook	5,921
Total	**30,000**

The distribution indicates that the number of records across the five preferred platforms is relatively balanced.

9. DATA VISUALIZATION AND INTERPRETATION

Four visualizations were created to explore the dataset. Different chart types were used to examine numerical distributions, category comparisons, relationships between variables and differences in distributions across groups.

9.1 HISTOGRAM – DISTRIBUTION OF DAILY SOCIAL MEDIA USAGE

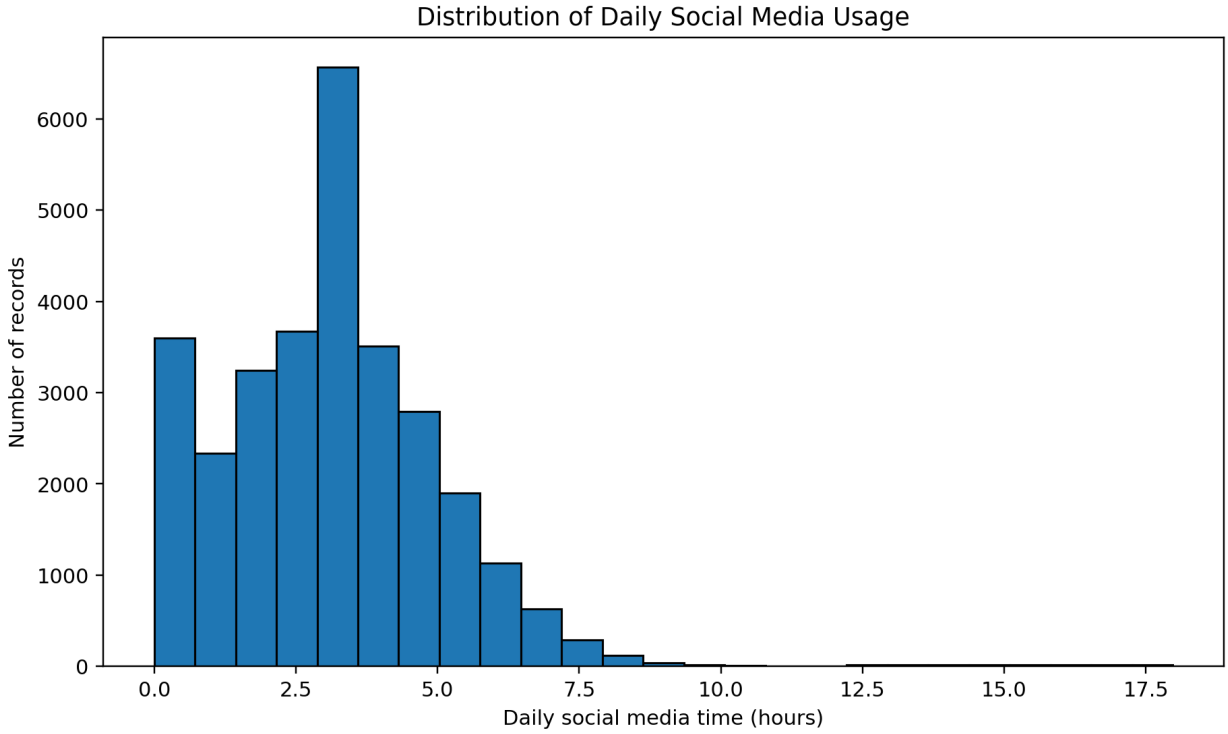


Figure 1: Distribution of Daily Social Media Usage

Interpretation:

The histogram represents the distribution of daily social media usage across the dataset.

The mean daily social media usage is approximately 3.11 hours, while the median is approximately 3.03 hours.

Most observations are concentrated in the lower-to-middle part of the displayed range, while a smaller number of records extend towards higher usage values.

The histogram helps understand how daily social media usage is distributed across the dataset.

9.2 BAR CHART – AVERAGE PRODUCTIVITY BY JOB TYPE

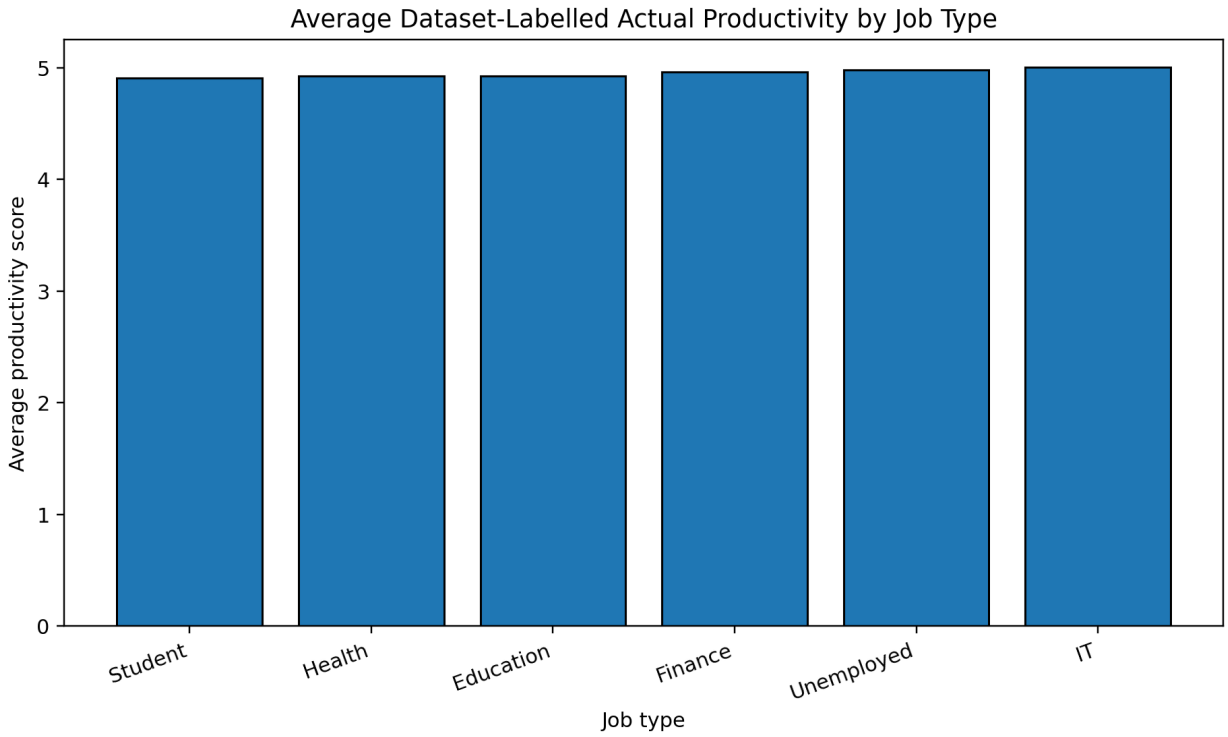


Figure 2: Average Dataset-Labelled Actual Productivity Score by Job Type

Interpretation:

The bar chart compares the average dataset-labelled actual productivity scores across six job categories.

The IT category has the highest average productivity score of approximately 5.003, while the Student category has the lowest average score of approximately 4.908.

The difference between the highest and lowest group averages is approximately 0.095 points.

The values are very close across all categories, indicating that there is no substantial difference in average productivity scores between the job categories in this dataset.

9.3 SCATTER PLOT – SOCIAL MEDIA USAGE VS PRODUCTIVITY

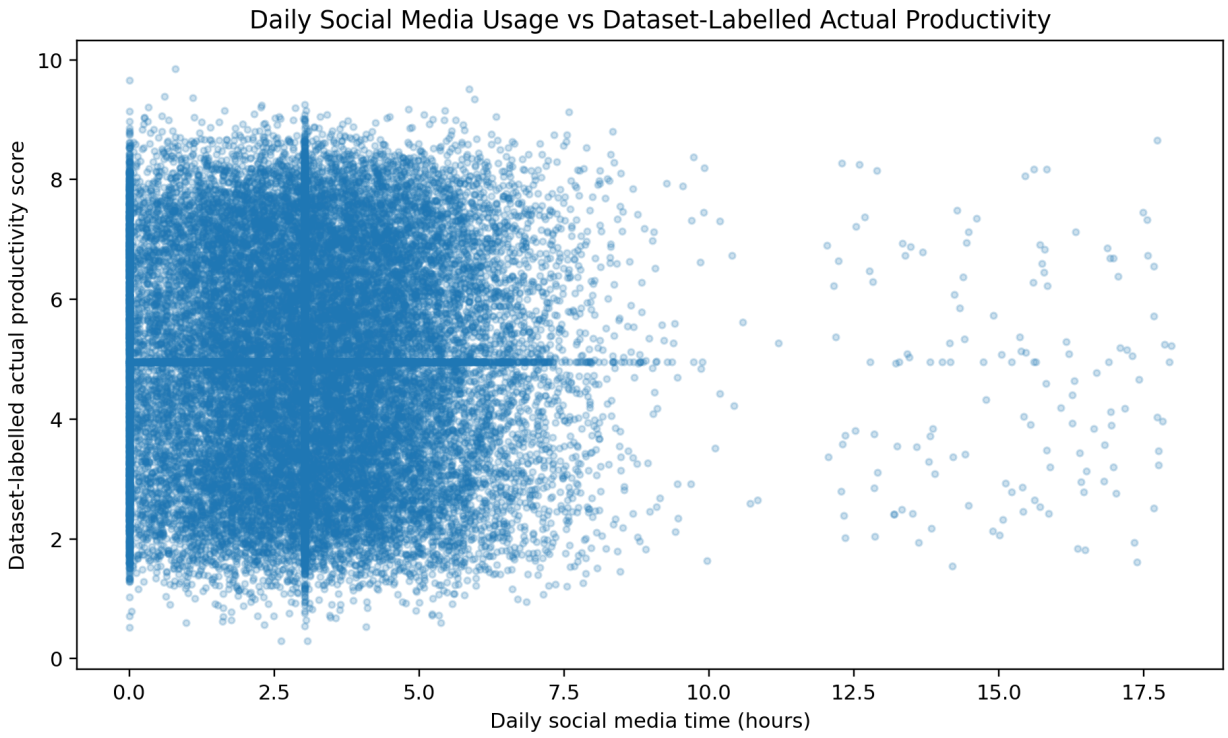


Figure 3: Relationship Between Daily Social Media Usage and Dataset-Labelled Actual Productivity

Interpretation:

The scatter plot examines the relationship between daily social media usage and the dataset-labelled actual productivity score.

Each point represents an observation in the dataset. The points are widely distributed, without a clear increasing or decreasing linear pattern.

The Pearson correlation coefficient is approximately −0.010.

This value is very close to zero, indicating an almost negligible linear association between the two variables in the supplied dataset.

The result does not establish a causal relationship between social media usage and productivity.

9.4 BOX PLOT – STRESS LEVELS ACROSS JOB TYPES

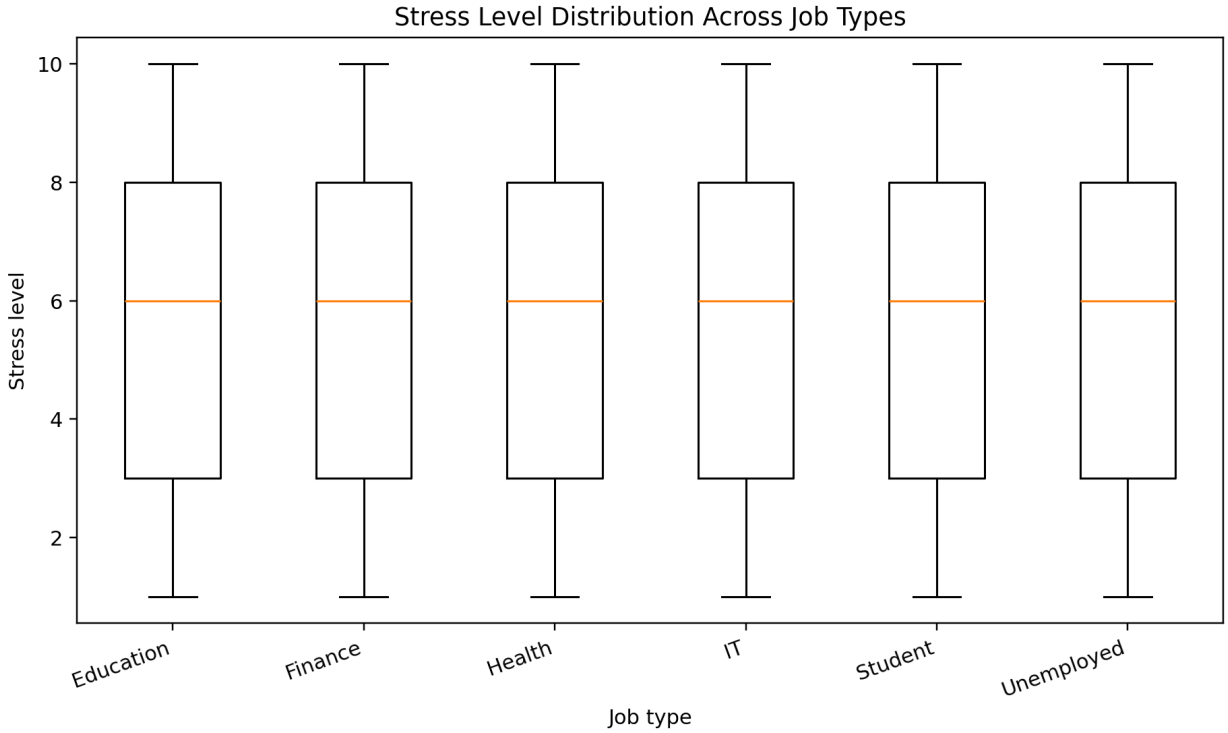


Figure 4: Distribution of Stress Levels Across Different Job Types

Interpretation:

The box plot compares the distribution of stress-level scores across the six job categories.

The first quartile, median and third quartile are 3, 6 and 8 respectively across the job categories.

The central distributions are therefore very similar across the different groups.

The visualization helps compare the spread and central tendency of stress levels and indicates that the differences between job categories are limited in this dataset.

10. KEY FINDINGS

Based on the exploratory data analysis and visualizations, the following findings were identified:

1. The average daily social media usage in the imputed dataset is approximately 3.11 hours.
2. The correlation between daily social media usage and the dataset-labelled actual productivity score is approximately −0.010, indicating an almost negligible linear relationship.
3. Average productivity scores are similar across the six job categories, with a difference of approximately 0.095 points between the highest and lowest group averages.
4. Stress-level quartiles are identical across the six job categories, with Q1 = 3, median = 6 and Q3 = 8.
5. The dataset does not contain a direct study-hours variable or a total daily screen-time variable.

6. The dataset is reported as synthetic or simulated, so the findings should be interpreted as exploratory patterns rather than verified observations from real individuals.

11. RECOMMENDATIONS

Based on the analysis, the following recommendations are suggested:

1. Individuals can monitor their screen-time patterns and identify whether particular usage periods interrupt planned work or study sessions.
2. Notification controls, focus modes and scheduled breaks can be used to reduce unnecessary digital interruptions during important tasks.
3. Productivity should not be evaluated using screen time alone, as it may also depend on task type, workload, sleep and individual circumstances.
4. Future studies should collect primary data directly from students, including daily study hours, total screen time, social media usage and self-reported productivity.
5. Future analysis should distinguish correlation from causation and avoid claiming that social media usage directly causes changes in productivity.

12. CONCLUSION

The analysis of the supplied dataset provides an overview of digital lifestyle and productivity-related variables.

The visualizations show the distribution of daily social media usage, small differences in average productivity scores across job categories, an almost negligible linear correlation between daily social media time and

the dataset-labelled actual productivity score, and similar central stress-level distributions across job types.

The results do not indicate a strong linear relationship between social media usage and the productivity score in this dataset.

However, the simulated nature of the dataset and the absence of direct study-hours and total-screen-time variables limit the conclusions. The project therefore demonstrates exploratory data analysis and visualization rather than proving how real students' digital habits affect academic productivity.

13. LIMITATIONS

The study has the following limitations:

1. The dataset is reported as synthetic or simulated and does not represent verified responses from 30,000 real individuals.
2. The dataset does not contain study hours. The work_hours_per_day attribute cannot be treated as a substitute for study hours.
3. The dataset contains screen_time_before_sleep rather than total daily screen time.
4. The dataset-labelled actual_productivity_score is not independently verified as an objective productivity measurement.
5. The dataset does not document a real-world collection period, sampling process or participant recruitment method.
6. Correlation and group comparisons do not establish cause-and-effect relationships.
7. The four figures use the median-imputed dataset before outlier removal and scaling.

14. FUTURE SCOPE

The project can be extended in the following ways:

1. Conduct a voluntary and anonymous survey of students using clearly defined measures of study hours, total screen time, social media usage and self-reported productivity.
2. Compare digital habits on weekdays and weekends.
3. Examine the relationship between sleep duration, notifications, focus-app usage and productivity measures.
4. Use a larger and more representative sample if the objective is to describe a wider student population.
5. Repeat the visualizations after preprocessing and document the exact dataset version used for each figure.

15. REFERENCES

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2. Younas, S., & Anande, T. J. (2026). *Digital wellbeing in the social media era: dual pathway analysis of emotional and behavioural distress indicators and public health utility.* medRxiv preprint. https://doi.org/10.64898/2026.10.09.26365101

APPENDIX A: TOOLS AND TECHNOLOGIES

Table 8: Tools and Technologies Used

Tool / Library	Purpose
Python	Programming language for data analysis
Google Colab / Jupyter Notebook	Environment for executing the analysis
pandas	Data loading, cleaning and manipulation
NumPy	Numerical operations
Matplotlib	Data visualization
Seaborn	Statistical visualization
scikit-learn	Numerical feature scaling using MinMaxScaler